

WM-D6C SERVO REPLACEMENT MODULE

DSR-1 | Revision 0.1 | Hardware + Firmware Module Data Sheet
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VARIANT A

USB-C Power Delivery

SHARED CORE

STM32G0B1KBU6 + Power

VARIANT B

Barrel Jack, Polarity-Agnostic

This document describes the complete hardware and firmware architecture of the WM-D6C Servo Replacement Module, an open source drop-in replacement for the capstan servo IC (CX20084), DC-DC boost converter (CP304), and DC power input jack (CN301) in the Sony Walkman WM-D6C professional cassette recorder. The module replaces these three ageing and increasingly unavailable components with a single compact PCB based on the STM32G0B1KBU6 microcontroller, improving upon the original design in speed stability, power supply protection, and long-term serviceability.

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Revision	0.1 — Initial Release
Status	Design Phase — Pending Bench Verification
Target Machine	Sony Walkman WM-D6C (all suffix revisions)
Primary MCU	STM32G0B1KBU6 — UFQFPN32, 64 MHz Cortex-M0+
Repository	github.com/[org]/wmd6c-servo (open source)
Hardware License	CERN Open Hardware Licence Strong v2 (CERN OHL-S v2)
Firmware License	MIT License
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1. Introduction and Background

1.1 The Sony WM-D6C

The Sony Walkman WM-D6C is a professional-grade portable cassette recorder produced by Sony from approximately 1984 through the early 1990s. Unlike consumer Walkman products, the WM-D6C was designed for broadcast journalists, field recordists, and professional audio applications, featuring full-size record and playback heads, Dolby B and C noise reduction, a precision capstan servo system, and a wow-and-flutter specification of 0.05% WRMS — exceptional for a handheld battery-powered device of its era.

The machine achieved its speed stability through a dedicated capstan servo IC, the Sony CX20084, which formed the heart of a phase-locked loop that compared the frequency-generator (FG) pulse train from the capstan motor against a 34.7 kHz crystal reference divided down by the MSM58141RS counter IC. The servo drove the base of a PNP transistor (Q601, type 2SB733) to modulate motor current and maintain precise tape transport speed.

The WM-D6C has acquired a devoted following among vintage audio enthusiasts, broadcast historians, and professional tape recordists who recognise its combination of portability, audio quality, and mechanical robustness. Surviving units are increasingly sought after, and the community supporting them is active and technically capable.

1.2 The Failure Mode This Module Addresses

The WM-D6C's DC power input connector (CN301, Sony part 1-507-749-00) is a non-standard 5.5mm barrel jack wired with negative polarity on the centre pin — the opposite of virtually every generic DC adapter sold since the 1980s. This was not unusual for Sony equipment of the period but has become a persistent hazard as original Sony AC adapters have been lost, discarded, or failed.

The consequence of connecting a standard centre-positive adapter to CN301 is catastrophic and immediate: the full adapter voltage is applied in reverse across the B+1 supply rail, which feeds the CX20084 servo IC directly with no protection diode or crowbar circuit. The IC fails instantly and irreversibly. The failure mode is electrically clean — the rest of the machine typically survives intact — but the CX20084 is a proprietary Sony ASIC that has been out of production for decades. Replacement ICs are scarce, expensive when found, and of uncertain provenance.

WARNING

The original CN301 power jack represents the single most common cause of electronic failure in surviving WM-D6C units. This module eliminates the hazard permanently by replacing CN301 with a correctly protected power input in both available variants.

1.3 Module Purpose and Scope

This module replaces three components simultaneously:

- The CX20084 capstan servo IC (IC601), replaced by digital servo firmware running on the STM32G0B1KBU6
- The CP304 DC-DC boost converter module (6V to 10.8V for the motor rail), replaced by an MT3608-based synchronous boost converter on the same PCB
- The CN301 DC power input jack, replaced either by a USB-C Power Delivery connector (Variant A) or by a polarity-agnostic protected barrel jack interface (Variant B)

All other WM-D6C circuitry is retained unchanged, including the motor drive transistors Q601, Q703, and Q704; the FG901 optical speed sensor; the speed trim potentiometers RV601, RV602, and RV603; the Speed Tune switch S601; and the Auto-Off board. The module interfaces with the existing machine through a keyed wire harness that connects to the original CP304 harness connector and to the solder pads of the removed CX20084.

1.4 Design Philosophy

Three principles govern every design decision in this module.

Cost and simplicity of installation drive adoption in the enthusiast community. A module that costs more than approximately thirty-five US dollars landed, or requires tools and skills beyond those needed for standard capacitor replacement, will not spread. Every component choice is evaluated against this constraint.

Open source hardware, firmware, and documentation ensure the community can maintain, adapt, and extend the design indefinitely. All design files are released under permissive open licences. The firmware is plain C compiled with standard open-source tools. The documentation is written for the competent amateur, not the specialist engineer.

Technical correctness without compromise means the digital servo loop must equal or exceed the original CX20084 in speed stability, and the power supply architecture must be more robust than the original in every measurable respect. The module does not cut corners in the signal chain in the name of simplicity.

2. Functional Overview

2.1 System Architecture

The module consists of a single two-layer PCB occupying the space of the original CP304 DC-DC board, with connection to the WM-D6C main board via the original harness and a short supplementary wire set. The board divides functionally into four zones.

The power input zone handles the variant-specific power source — either a USB-C PD charger negotiated to 9V (Variant A) or a barrel jack adapter of either polarity (Variant B) — and produces the regulated internal supply rails required by the module and the WM-D6C machine.

The power management zone contains the MT3608 synchronous boost converter that produces the 10.8V B+3 motor rail previously generated by CP304, and the MCP1700 low-dropout regulator that produces the 3.3V supply for the STM32. The regulated 6V B+1 rail is sourced directly from the power input zone and fed to the WM-D6C main board via the harness, replacing the battery or original DC input.

The microcontroller zone contains the STM32G0B1KBU6 and its support components: decoupling capacitors, the FG signal conditioning network, the motor drive output stage, and the ADC input dividers. All servo loop processing, USB communication, and power delivery negotiation occurs within the STM32.

The interface zone contains the WM-D6C machine connector J1, which is an eight-position 1.25mm JST connector carrying all signals between the module and the original machine circuitry.

2.2 Variant Differentiation

The two variants share an identical PCB gerber, BOM structure, and STM32 firmware. They differ only in which components are populated in the power input zone and which connector or port is provided for DC power.

Parameter	Variant A — USB-C PD	Variant B — Barrel Jack
Power Input	USB-C Type C receptacle, panel mount or flex tail to CN301 opening	5.5mm barrel jack, CN301 retained in original panel opening
Input Voltage	9V negotiated via USB PD (IP2721 trigger IC)	4.5V to 9V, either polarity accepted
PD Negotiation	IP2721 hardware PD trigger — no firmware required	Not applicable
Polarity Protection	Inherent in USB-C physical format	LTC4359 ideal diode bridge controller, four-MOSFET topology
Overcurrent	USB PD source current limiting	500mA polyfuse (MF-R050 or equivalent, resettable)
Overvoltage	PD contract voltage capped at 9V by IP2721	SMBJ7.0A TVS diode, 7V standoff, transient clamp
Populated Zone	Zone 3 (USB-C components)	Zone 4 (barrel jack protection components)
DNP Zone	Zone 4 (barrel jack)	Zone 3 (USB-C)
CN301 Disposition	Removed, opening modified or fitted with flex panel connector	Retained in original panel position

2.3 Power Rails

The module generates or passes through three power rails to the WM-D6C machine and its own internal circuits.

Rail	Consumers	Regulation	Current
B+1 (6.0V)	Audio circuits, logic, servo reference, Auto-Off board	5.85V to 6.15V regulated	600mA continuous

B+3 (10.8V)	Capstan motor M901 via Q703/Q704 switching	10.5V to 11.0V	500mA peak during acceleration
VDD (3.3V)	STM32G0B1KBU6, VDDIO2, USB PHY	3.27V to 3.33V	250mA

3. Microcontroller — STM32G0B1KBU6

3.1 Device Selection Rationale

The STM32G0B1KBU6 was selected as the central processing element of this module after evaluation against the functional requirements of the application. The device is a 32-bit Arm Cortex-M0+ microcontroller produced by STMicroelectronics in the STM32G0 family, operating at up to 64 MHz with 128 KB of flash memory and 144 KB of SRAM.

Several features of this specific device make it uniquely well-suited to this application. The built-in USB 2.0 Full Speed device controller with crystal-less operation eliminates the need for an external USB bridge chip or crystal oscillator — the 48 MHz clock required by the USB PHY is generated from the internal HSI48 oscillator with automatic frequency trimming via the USB Start-of-Frame signal, requiring no external components for USB operation. The built-in USB Type-C Power Delivery controller (UCPD1) on pins PA8 and PB15 handles CC line detection and PD message exchange without external hardware in Variant A. The 12-bit DAC provides a clean analog motor drive output without the need for an external digital-to-analog converter. The 12-bit ADC with three simultaneously available input channels reads all three speed trim potentiometers in a single scan sequence.

The UFQFPN32 package — a 5 × 5 mm, 0.5 mm pitch 32-pad QFN with an exposed thermal pad — is the smallest package in the STM32G0B1 family that carries all the peripherals required for this application. It is within the standard process capability of JLCPCB for both bare board fabrication and SMT assembly, and it accepts standard reflow soldering in small-batch community builds.

3.2 Key Specifications

Full Part Number	STM32G0B1KBU6
Package	UFQFPN32 — 5.0 × 5.0 mm, 0.5 mm pitch QFN, exposed VSS pad
Core	Arm Cortex-M0+, 32-bit, single-cycle multiply
Core Clock	64 MHz maximum (HSI16 × PLL)
Flash Memory	128 KB with read-while-write capability
SRAM	144 KB (20 KB reserved for CCM, 122 KB general purpose)
Operating Voltage	2.0V to 3.6V (VDD), 1.71V to 3.6V (VDDIO2)
Operating Temperature	−40°C to +85°C ambient (−40°C to +105°C junction)
USB	Full Speed 12 Mbit/s, USB 2.0 compliant, crystal-less via SOF trimming
USB PD Controller	UCPD1 and UCPD2, USB Type-C Rev 1.2, USB PD Rev 3.0
Timers	TIM1 (advanced), TIM2/3/14/15/16/17 (general), LPTIM1/2
ADC	12-bit, up to 1 Msps, 15 channels including internal VREF and temperature
DAC	12-bit, two channels, buffer output, sine wave generation capable
DMA	5-channel DMA controller
Boot Time to Servo Loop	Approximately 2 ms from power-on
Thermal Resistance (junction-ambient)	35°C/W in UFQFPN32 package

3.3 Clock Configuration

The STM32G0B1KBU6 in this application uses no external crystal oscillator. The primary system clock is derived from the internal HSI16 RC oscillator (16 MHz, ±1% accuracy at 25°C) multiplied by the internal PLL to produce the 64 MHz system clock. This configuration requires no external components and is immune to crystal cracking or aging — a relevant concern in a device subject to decades of storage and handling.

The 48 MHz clock required by the USB peripheral is sourced from the internal HSI48 oscillator, which is automatically trimmed against the USB data stream via the Clock Recovery System (CRS) peripheral using the USB Start-of-Frame packet. This approach, documented in ST Application Note AN4879, achieves better than 0.25% USB clock accuracy in normal operation without any external crystal.

NOTE

The STM32G0B1Kx datasheet (DS13560 Rev 6, Section 3.24) explicitly confirms crystal-less USB device operation via internal 48 MHz oscillator with SOF synchronisation. This eliminates a component, a failure mode, and a layout constraint simultaneously.

3.4 Pin Assignment

All pin assignments are confirmed against the STM32G0B1KBU UFQFPN32 pinout in DS13560 Rev 6, Figure 4 and Table 12.

Pin	Function	Direction	Notes
PA0 / TIM2_CH1	FG pulse input capture	Input	FG_IN from FG901 via level shift network
PA1 / ADC_IN1	Speed base trim	Analog input	RV601 wiper (47kΩ cermet)
PA2 / ADC_IN2	Speed tune slider	Analog input	RV602 wiper (20kΩ carbon)
PA3 / ADC_IN3	Speed range trim	Analog input	RV603 wiper (47kΩ cermet)
PA4 / DAC1_OUT1	Motor drive analog output	Analog output	DAC output → motor drive stage
PA5 / GPIO	Motor enable monitor	Input	IC601 pin 7 net via 10k/22k divider
PA6 / TIM3_CH1	PWM motor drive (Option B)	Output	PWM → RC filter → NPN level shift
PA7 / GPIO	Speed Tune switch	Input	S601 slide switch
PA8 / UCPD1_CC1	USB PD CC1	UCPD I/O	Variant A: CC1 line to USB-C connector
PA9 / USART1_TX	Debug UART transmit	Output	Test pad J2 only, optional
PA11 / USB_DM	USB D− data line	USB I/O	To USB-C D− via ESD clamp
PA12 / USB_DP	USB D+ data line	USB I/O	To USB-C D+ via ESD clamp
PA13 / SWDIO	SWD data	Debug	SWD programming header J1
PA14 / SWDCLK	SWD clock	Debug	SWD programming header J1
PB15 / UCPD1_CC2	USB PD CC2	UCPD I/O	Variant A: CC2 line to USB-C connector
VDD (pins 1, 32)	Digital supply	Power	3.3V from MCP1700, 100nF × 4 decoupling
VDDA (pin 5)	Analog supply	Power	3.3V, 1μF + 100nF decoupling
VDDIO2 (pin 18)	VDDIO2 domain supply	Power	3.3V, 100nF + 4.7μF decoupling
VSS / Exposed pad	Ground	Power	Board GND plane, via array to GND pour

4. Capstan Servo Loop

4.1 Theory of Operation

The WM-D6C capstan servo system maintains constant tape speed by regulating the rotational speed of the capstan shaft. A slotted optical disc attached to the capstan motor shaft interrupts the beam of the FG901 GP2S22AB optical sensor, producing a pulse train whose frequency is proportional to motor speed. The servo controller measures this frequency, compares it to a target value representing the correct tape speed, and adjusts the motor drive current accordingly.

In the original CX20084 implementation this comparison was performed in the analog domain: a phase-locked loop compared the divided FG pulse stream against the divided output of the 34.7 kHz X701 crystal oscillator, generating an error voltage that modulated the motor drive output through an analog integration network. The time constants of this network determined the servo bandwidth and settling behaviour.

The STM32 implementation performs the equivalent function entirely in the digital domain. The TIM2 input capture hardware timestamps each rising edge of the FG pulse train with single-clock-cycle precision. The ISR computes the period between successive captures, compares it to a pre-calculated target period, and applies a proportional-integral control law to generate a new motor drive value. This is written to either the 12-bit DAC (Option A, direct analog output) or to the TIM3 PWM register followed by an RC filter (Option B, for cases where the motor drive circuit requires a voltage above 3.3V).

The digital implementation offers measurable advantages over the original analog circuit. The target period is a fixed integer value computed from the known FG pulse rate at correct tape speed — it does not drift with temperature, component aging, or supply voltage variation. The PI coefficients are stored in flash and can be updated via USB CDC without any hardware modification. The integral accumulator prevents sustained steady-state speed error regardless of motor load variation across the tape reel.

4.2 FG Signal Conditioning

The FG901 GP2S22AB optical sensor is an open-collector device. Its output is pulled up to a voltage on the WM-D6C main board that is approximately equal to the B+1 rail voltage — nominally 6V from batteries, possibly somewhat lower under load. The STM32G0B1KBU6 PA0 pin has an absolute maximum input voltage of $V_{DD} + 0.3V$, which at 3.3V supply means 3.6V. The FG signal must therefore be attenuated before entering the STM32.

The conditioning network consists of a resistive voltage divider ($R3 = 10k\Omega$ series, $R4 = 22k\Omega$ to ground) that attenuates a 5V FG signal swing to approximately 3.44V. A pair of BAT54 Schottky diodes clamp the conditioned signal to V_{DD} (3.3V) and GND respectively, providing hard clamping against transients and the case where the FG swing is higher than anticipated. A 10pF C0G capacitor to ground filters high-frequency noise introduced by the slotted disc mechanism.

WARNING

The exact FG signal voltage swing must be measured on the specific unit during initial bench bring-up before connecting to the STM32. The $R3/R4$ divider values listed above assume a 5V swing. If the actual swing is different, $R4$ must be recalculated to ensure the conditioned signal does not exceed 3.3V under any condition.

4.3 PI Control Algorithm

The control law is a standard discrete-time proportional-integral controller operating in the period domain rather than the frequency domain. Period domain operation has the advantage that the measured quantity (timer capture ticks) is already in the correct form for arithmetic without division.

```
// Called from TIM2_IRQHandler on each FG rising edge
// All arithmetic in 32-bit integer — no floating point on Cortex-M0+

static int32_t integral = 0;
static uint32_t last_capture = 0;

void TIM2_IRQHandler(void) {
```



```

uint32_t now      = TIM2->CCR1;
uint32_t period   = now - last_capture;
last_capture      = now;

// Error: positive = running too slow (period too long)
// PNP Q601 convention: more drive = lower base = lower DAC output
int32_t error = (int32_t)period - (int32_t)target_period;

// Integral with windup clamp
integral += error;
if (integral > INTEGRAL_LIMIT) integral = INTEGRAL_LIMIT;
if (integral < -INTEGRAL_LIMIT) integral = -INTEGRAL_LIMIT;

// Control output (Q16 fixed-point Kp and Ki)
int32_t output = DAC_CENTER
    - ((KP_Q16 * error)    >> 16)
    - ((KI_Q16 * integral) >> 16);

// Clamp to DAC range 0..4095
if (output < 0)      output = 0;
if (output > 4095)   output = 4095;

DAC1->DHR12R1 = (uint32_t)output;
TIM2->SR &= ~TIM_SR_CC1IF; // clear interrupt flag
}

```

The initial PI coefficients are set by bench tuning against a calibrated test tape. Starting values are $K_p = 0.15$ and $K_i = 0.008$ in floating-point equivalents before Q16 conversion; these are stored as Q16 fixed-point integers (9830 and 524 respectively) in a settings block in the final 4 KB sector of flash memory. The USB CDC command interface allows these values to be adjusted in real time and saved to flash without reflashing the firmware.

4.4 Motor Drive Output Stage

Two output stage configurations are supported. The correct configuration for a given unit must be selected based on bench measurement of the Q601 base voltage range during normal playback. This measurement determines whether the motor drive circuit can be driven directly from the STM32 3.3V DAC output or requires a level-shifting stage.

Option A — Direct DAC Drive

Used when the Q601 base operating voltage is within the range 0V to 3.3V. The DAC output on PA4 drives the base of Q601 through a 10k Ω series resistor R5. A 100k Ω resistor R6 pulls the base to VDD (3.3V), which holds Q601 off during boot before the servo loop initialises — the PNP transistor requires a base voltage lower than its emitter to conduct. R6 ensures the motor is off at power-on regardless of DAC output state.

Option B — NPN Level-Shift Drive

Used when the Q601 base operating voltage is above 3.3V, which occurs when the B+1 rail voltage drives the emitter of Q601 at 6V and the base must be controlled in the range 5.2V to 6V. An MMBT3904 NPN transistor Q_LS is interposed between the STM32 and Q601. The STM32 PA6 TIM3 PWM output drives the base of Q_LS through a 1k Ω resistor R7. An RC filter ($C_8 = 100\text{nF}$ to GND) reconstructs the analog control voltage from the PWM signal. The collector of Q_LS connects to the base of Q601 through a 10k Ω resistor R8, with a 100k Ω pullup R9 from the B+1 rail to the Q601 base ensuring motor-off at boot.

NOTE

Option A is populated on all boards by default. If bench measurement of Q601 base voltage indicates Option B is required, R5/R6 are replaced with DNP (Do Not Populate) designators and the Option B components Q_LS, R7, R8, R9, and C8 are installed. The firmware detects the configuration via a solder bridge JP1 and switches between DAC and PWM output modes accordingly.

4.5 Speed Reference Potentiometers

Three potentiometers from the original WM-D6C circuit are read by the STM32 ADC and incorporated into the servo reference calculation.

Potentiometer	Type	Function
RV601 → PA1 / ADC_IN1	47kΩ cermet trimmer	Base speed calibration trim. Sets the fundamental target period offset. Factory-set during original servo calibration, should not require adjustment after module installation if the machine was previously calibrated correctly.
RV602 → PA2 / ADC_IN2	20kΩ carbon slider, front-panel Speed Tune control	User-accessible speed adjustment. The STM32 reads this continuously and adds a signed offset to the target period in real time. Adjustment range equivalent to the original CX20084 implementation.
RV603 → PA3 / ADC_IN3	47kΩ cermet trimmer	Speed range scalar. Sets the sensitivity of the RV602 Speed Tune control, determining how much frequency offset is produced by full travel of the slider.

The ADC scan runs in DMA mode, continuously updating the three pot readings in a background array. The servo ISR reads this array each execution cycle. Wiper voltages must be verified on the bench before connecting — if any wiper can exceed 3.3V under any operating condition, voltage dividers must be added before the ADC inputs.

5. Power Supply Architecture

5.1 Shared Power Management

5.1.1 MT3608 Boost Converter — 6V to 10.8V (B+3 Motor Rail)

The MT3608 is a synchronous step-up switching regulator in a SOT-23-6 package, replacing the CP304 potted boost module. It accepts the 6V B+1 input and produces the 10.8V B+3 rail that supplies the capstan motor M901 via the existing Q703/Q704 motor switching transistors. The output voltage is set by a resistor divider on the feedback pin. A 4.7 μ H inductor and SS14 Schottky rectifier diode complete the boost topology.

The MT3608 operates at a switching frequency of approximately 1.2 MHz, which is well above the audio band and above any mechanical resonance in the motor assembly. The choice of switching frequency is deliberate: at 1.2 MHz the required inductor is small (4.7 μ H in a 4 $\times$ 4mm package), switching noise is far removed from audible frequencies, and the output ripple at 10.8V with a 22 μ F output capacitor is under 50mV peak-to-peak — comparable to or better than the original CP304 performance.

5.1.2 MCP1700 LDO — 6V to 3.3V (MCU Supply)

The MCP1700T-3302E/TT is a 250mA low-dropout linear regulator in a SOT-23-3 package. It produces the 3.3V supply for the STM32G0B1KBU6 and all associated 3.3V logic. The dropout voltage of 178mV at 100mA means regulation is maintained down to approximately 3.5V input — the module continues to operate correctly until the B+1 rail drops below 3.5V, well below the 4.5V minimum that makes the WM-D6C machine itself operate marginally.

Input and output decoupling follows the MCP1700 datasheet recommendation: 1 μ F ceramic on the input, 1 μ F ceramic on the output, supplemented by 10 μ F electrolytic on both rails for bulk storage. Four additional 100nF 0402 ceramics are placed at the VDD pins of the STM32 on the board, one per pin pair.

5.2 Variant A — USB-C Power Delivery Input

5.2.1 IP2721 USB PD Trigger

The IP2721 is a single-chip USB Power Delivery sink controller in a SOT-23-6 package that negotiates a fixed voltage PDO (Power Data Object) from a USB PD source without requiring any firmware. A single resistor on the CFG pin programs the requested voltage. For this application the resistor is chosen to request 9V, which is available from virtually all USB PD chargers including phone chargers, laptop chargers, and portable power banks.

Once a 9V PD contract is established, the IP2721 closes its internal power path FET and presents 9V on VBUS to the downstream circuitry. If the connected source does not support PD or cannot supply 9V, the IP2721 falls back to 5V — this is insufficient to power the WM-D6C correctly and the module will not operate, protecting the machine from under-voltage operation. A status LED on the IP2721's PG (power good) output provides a simple power-on indicator.

5.2.2 TPS62xx Buck Regulator — 9V to 6V

A TPS62xx-series synchronous buck converter steps the 9V PD output down to the 6.0V B+1 rail required by the WM-D6C machine. The output voltage is set by an external resistor divider to precisely 6.0V. The device operates at a fixed switching frequency of 1.5 MHz and requires only an inductor, two capacitors, and the feedback resistors as external components, fitting the package footprint in a SOT-23-5.

5.2.3 USB Data Path

The USB-C connector carries both the power delivery CC lines and the USB 2.0 data lines D+ and D-. The CC lines connect to PA8 (CC1) and PB15 (CC2) of the STM32, which handles PD contract monitoring via the UCPD1 peripheral. The D+ and D- lines connect to PA12 and PA11 respectively through a USBLC6-2SC6Y ESD protection array, which provides 15kV ESD immunity per IEC 61000-4-2 on both data lines.

5.3 Variant B — Barrel Jack Input

5.3.1 Polarity-Agnostic Input Stage

The fundamental design requirement of Variant B is that the module must operate correctly when powered by the original Sony AC-D4M adapter (negative centre, 6V, 400mA) or by any generic positive-centre 6V adapter — without the installer needing to identify the polarity of their specific adapter.

This is achieved using the LTC4359 ideal diode bridge controller driving four external P-channel MOSFETs in a full-wave bridge configuration. Regardless of the polarity presented at the barrel jack, the bridge output is always the correct polarity, with a voltage drop of less than 50mV at rated current. This is in contrast to a conventional diode bridge rectifier, which would drop approximately 1.2V across two diodes in series — unacceptable given a 6V nominal input and 5.85V minimum machine requirement.

5.3.2 Protection Stack

Following the polarity bridge, a series of protection devices prevents damage from the fault conditions most likely to affect a 30-plus-year-old machine used with a variety of adapters.

- Resettable polyfuse (500mA hold, 1.5A trip): limits current during fault conditions. Resets automatically when fault is cleared. Prevents the adapter overload that could result from a wiring fault in the machine.
- SMBJ7.0A TVS diode: 7V standoff bidirectional transient suppressor. Clamps any voltage spike above 7V to ground. Protects against motor back-EMF transients that can propagate upstream through the motor rail switching circuitry.
- 220 μ F / 10V electrolytic input bulk capacitor: provides reservoir storage to ride through current demands during motor acceleration without drooping the input voltage.
- 100nF ceramic input bypass: high-frequency decoupling at the power entry point.

6. USB Interface and Firmware

6.1 USB CDC Virtual COM Port

The STM32G0B1KBU6 implements a USB CDC (Communications Device Class) virtual COM port that appears as a standard serial port on any computer running Windows, macOS, or Linux without driver installation. This interface provides two functions: live servo tuning during development and initial installation, and firmware updates using the STM32 USB DFU (Device Firmware Upgrade) bootloader.

The USB CDC interface runs in the main loop background at the lowest software priority. Because the servo loop executes entirely within hardware interrupt service routines, USB servicing does not introduce any jitter or latency into the control loop. The interrupt preemption hierarchy ensures the TIM2 capture ISR always runs within 500 nanoseconds of the FG edge regardless of USB activity.

6.2 Live Tuning Command Interface

The following single-character commands are processed by the USB CDC receive handler. Commands take effect immediately on the next servo ISR execution after receipt.

Command	Function	Detail
p+ / p-	Adjust proportional gain Kp	Increment or decrement Kp by 0.01 (Q16: ± 655). Affects the immediate response to speed error.
i+ / i-	Adjust integral gain Ki	Increment or decrement Ki by 0.001 (Q16: ± 65). Affects steady-state error elimination rate.
t	Print telemetry snapshot	Sends a formatted text block: FG period (ticks and Hz), target period, error, integral accumulator value, DAC/PWM output, and all three ADC pot readings.
T	Enable continuous telemetry	Sends telemetry on every FG pulse. Useful with a terminal plotting tool. Disable with any other character.
s	Save constants to flash	Writes current Kp, Ki, and target period to the last 4KB flash sector. Survives power cycling.
r	Reset to defaults	Restores factory Kp, Ki, and target period from compiled-in constants. Does not overwrite flash until 's' is sent.
f+ / f-	Adjust target period	Fine-tunes the master speed reference. Each step changes the target period by 1 tick (64 MHz).

6.3 Firmware Update via USB DFU

The STM32G0B1KBU6 contains a factory-programmed USB DFU bootloader in system memory. Entering DFU mode requires holding the BOOT0 pad (exposed as test point TP1 on the PCB) to VDD while applying power. The device then enumerates as a DFU device and accepts firmware in .bin format.

The recommended update tool for Windows is STM32CubeProgrammer (free, from ST). On macOS and Linux, dfu-util (open source, available via Homebrew and most package managers) works without any additional drivers. The workflow for an end user with no electronics background is: hold TP1 while connecting USB-C, drag the firmware .bin file in STM32CubeProgrammer, disconnect and reconnect. The module reboots into the new firmware immediately.

TIP

Pre-built firmware .bin files for each tagged release are stored in firmware/releases/ in the project repository. End users never need to install a compiler, IDE, or development tools to update their module.

6.4 Firmware Architecture

The firmware is written in minimal C, compiled with arm-none-eabi-gcc at optimisation level -O2. No HAL, no RTOS, no middleware. The firmware maps directly to STM32G0B1 peripheral registers through volatile pointer dereferences. There are no driver layers, callback chains, or abstraction frameworks between the control algorithm and the hardware.

The source is structured as follows. A single startup file in assembly initialises the stack, copies initialised data from flash to SRAM, zeroes the BSS segment, and calls main(). Clock initialisation configures HSI16, the PLL, and the peripheral bus clocks. Peripheral initialisation files (one per peripheral: tim2.c, dac.c, adc.c, usb.c, ucpd.c) configure each peripheral to its operating mode and enable interrupts. The servo algorithm lives entirely in servo.c and is self-contained — it has no dependencies on any other module except the shared constants defined in config.h.

```
// config.h — all tunable constants in one place
#define SYSCLK_HZ      64000000UL
#define FG_TARGET_HZ   2500UL           // measured on bench
#define TARGET_PERIOD  (SYSCLK_HZ / FG_TARGET_HZ) // = 25600 ticks
#define KP_Q16          9830            // 0.15 in Q16 fixed-point
#define KI_Q16          524             // 0.008 in Q16 fixed-point
#define INTEGRAL_LIMIT (TARGET_PERIOD * 50) // anti-windup clamp
#define DAC_CENTER     2048            // 12-bit midscale
#define DAC_MIN        512
#define DAC_MAX        3584
```

The total compiled firmware image occupies approximately 28 KB of the 128 KB flash, leaving 100 KB unused. RAM consumption is under 2 KB including stack. The remaining resources are available for future feature additions without hardware changes.

7. WM-D6C Machine Interface

7.1 Interface Connector J1

All signals between the module and the WM-D6C main board are carried by a single 8-position 1.25mm JST PH connector J1. The connector is keyed and cannot be inserted in the wrong orientation. A pre-assembled wire harness is included with the module, colour-coded to match the WM-D6C service manual wiring convention.

J1 Pin	Net Name	Direction	Connection
1	FG_RAW	Input	FG901 optical sensor output — connect to FG901 collector, pull-up on main board
2	Q601_BASE	Output	Motor drive to Q601 (2SB733) base — connect to CX20084 pin 15 solder pad
3	MOTOR_EN	Input	Motor enable from Auto-Off board — IC601 pin 7 net, 4.4V during playback
4	RV601_WIPER	Analog input	Base speed trim potentiometer wiper — 47kΩ cermet RV601
5	RV602_WIPER	Analog input	Speed Tune slider wiper — 20kΩ carbon RV602, front panel control
6	RV603_WIPER	Analog input	Speed range trim potentiometer wiper — 47kΩ cermet RV603
7	B+1 / VBATT	Power in/out	6V regulated supply rail — feeds machine and module when on batteries
8	GND	Power	WM-D6C chassis ground — connected to module GND plane

7.2 Signal Voltage Compatibility

All signals at J1 originate from or are consumed by circuits operating at B+1 voltage (nominally 6V), which is incompatible with the STM32G0B1KBU6 3.3V logic levels without conditioning. The following conditioning is applied to each signal.

Signal	Conditioning	Notes
FG_RAW (pin 1)	R3 10kΩ + R4 22kΩ voltage divider, BAT54 Schottky clamp diodes D2/D3 to 3.3V and GND, 10pF C0G filter cap C7	Actual R3/R4 values require verification of FG901 swing voltage on the specific unit. Default values assume 5V swing → 3.44V conditioned.
Q601_BASE (pin 2)	Option A: PA4 DAC → R5 10kΩ → Q601 base, R6 100kΩ pullup to 3.3V. Option B: TIM3 PWM → R7 1kΩ → C8 100nF → MMBT3904 Q_LS → R8 10kΩ → Q601 base	Option selection based on measured Q601 base voltage. If base operates below 3.3V, Option A. If above, Option B.
MOTOR_EN (pin 3)	R10 10kΩ + R11 22kΩ voltage divider to PA5 GPIO input	$4.4V \times 22/(10+22) = 3.03V$ at PA5. Confirmed safe.
RV601/602/603 (pins 4-6)	Direct connection to ADC pins with BAT54 clamp diodes to 3.3V and GND	Wiper voltages must be measured. If any exceeds 3.3V under any condition, voltage dividers required before ADC.

7.3 Components Removed from WM-D6C

The following original WM-D6C components are removed as part of the module installation. Removal is permanent — these components are not reinstalled if the module is later removed.

Component	Location / Type	Disposition
CX20084 (IC601)	Capstan servo IC, Auto-Off board position E-4	Desoldered from main board. All pins removed. Solder pads cleaned. Pins 13, 14, 15, and GND used for harness connection.
X701 crystal	34.7 kHz servo reference oscillator, Auto-Off board	Desoldered. No longer needed — STM32 uses internal oscillators.
IC701 MSM58141RS	Crystal divider and phase comparator, Auto-Off board	Desoldered. No longer needed — function replaced by STM32 firmware.
CP304 module	DC-DC boost converter module, DC-DC board cavity	Unplugged from harness connector. Module removed. New PCB installs in same cavity using same harness connector.
CN301 (Variant A only)	DC power input barrel jack, lower panel	Desoldered. Panel opening modified (filed oval) to accept USB-C panel connector, or USB-C flex tail fitted.

7.4 Components Retained and Reused

The following original WM-D6C components remain in place and are driven or read by the module.

Component	Type	Notes
Q601 (2SB733)	PNP motor current control transistor	Driven by module output stage. Test for continuity (diode mode) before installation — may have been damaged in same reverse polarity event as CX20084.
Q703 / Q704	Motor switching transistors	Untouched. Mode selection logic drives these transistors as in the original circuit.
FG901 (GP2S22AB)	Optical speed sensor	Read by module via J1 pin 1. Clean slotted disc with IPA during mechanical service.
RV601 / RV602 / RV603	Speed trim and tune potentiometers	Read by module ADC via J1 pins 4/5/6. No electrical change to potentiometer circuit.
S601	Speed Tune enable switch	Read by PA7 GPIO via J1 harness. No electrical change.
CN301 (Variant B only)	Barrel jack, retained in panel	Connected to Variant B power input zone. Original Jack, now protected against reverse polarity.

8. PCB Design

8.1 Board Specification

Target Dimensions	To be confirmed by measurement of CP304 cavity — estimated 30 × 22mm
Layer Count	2-layer (top signal + power, bottom ground plane)
Material	FR-4, Tg 130°C minimum
Copper Weight	1 oz (35µm) both layers
Surface Finish	ENIG (Electroless Nickel Immersion Gold)
Solder Mask	Green, both sides
Silkscreen	White, top side, component designators and reference marks
Board House	JLCPCB standard 2-layer process (or equivalent)
Design Rule	6/6 mil (0.15mm trace / 0.15mm space) minimum
Via Size	0.3mm drill, 0.6mm annular ring minimum
QR Code	Printed in silkscreen linking to installation guide URL

8.2 Component Package Summary

All active components are surface mount. Passive components (resistors, capacitors) use 0402 imperial package unless otherwise noted. The following table lists the package for each IC.

Component	Package	Notes
STM32G0B1KBU6	UFQFPN32	5.0 × 5.0 mm, 0.5mm pitch, 32-pad QFN, exposed VSS pad
MT3608	SOT-23-6	2.9 × 1.6 mm
MCP1700T-3302E/TT	SOT-23-3	2.9 × 1.6 mm
IP2721 (Variant A)	SOT-23-6	2.9 × 1.6 mm
TPS62xx buck (Variant A)	SOT-23-5	2.9 × 1.6 mm
USBLC6-2SC6Y (Variant A)	SOT-23-6	2.9 × 1.6 mm
LTC4359 (Variant B)	SOT-23-5	2.9 × 1.6 mm
MMBT3904 Q_LS (Option B)	SOT-23-3	2.9 × 1.6 mm
BAT54 clamp diodes	SOD-123	3.8 × 1.5 mm
SS14 boost rectifier	SMA	5.0 × 2.7 mm
4.7µH boost inductor	4040 (4.0 × 4.0 mm)	Maximum 3.0 mm height
SMBJ7.0A TVS (Variant B)	SMA / SMB	3.6 × 2.8 mm
USB-C receptacle (Variant A)	Mid-mount SMD	8.94 × 7.36 mm + PCB tail

8.3 Layout Guidelines

The following layout rules apply to the board design and are critical for reliable operation.

Power Conversion

- MT3608 switching loop (inductor, input capacitor, output capacitor, diode) must be routed with minimum loop area to reduce switching noise radiation.
- Boost inductor L1 placed within 3mm of MT3608 SW pin. Output capacitors placed within 3mm of MT3608 output.
- MCP1700 input and output decoupling capacitors placed within 2mm of regulator pins.

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- Exposed VSS pad connected to GND pour via 2×2 array of 0.3mm vias. Top-side GND pour under device tented on top, plugged or tented on bottom to prevent solder wicking during reflow.
- VDDIO2 decoupling (100nF + 4.7μF) placed within 2mm of pin 18.
- VDDA decoupling (100nF + 1μF) placed within 2mm of pin 5.
- Four VDD decoupling 100nF capacitors placed as close as possible to the four VDD pins (pins 1, 16, 17, 32).

Signal Integrity

- USB D+ and D− routed as a matched-length differential pair with 90Ω differential impedance. Traces kept under 25mm from connector to PA11/PA12.
- FG signal conditioning network (R3, R4, D2, D3, C7) placed close to J1 pin 1 and away from switching converter.
- ADC input traces (J1 pins 4/5/6 to PA1/PA2/PA3) kept away from the MT3608 switching node.

9. Installation Procedure

The following procedure applies to installation of the module in a Sony WM-D6C. The procedure assumes a machine that has already received a full mechanical service including belt replacement, pinch roller replacement, FG disc cleaning, and capstan bearing lubrication. Mechanical service should always precede electronic diagnosis and module installation.

WARNING

Before opening the machine, verify the original fault using a calibrated test tape and confirm the machine exhibits the known symptom (playback speed too fast, uncontrolled). Complete the mechanical service and retest. Only proceed with module installation if the fault persists after mechanical service.

9.1 Tools Required

- Phillips #00 and #000 screwdrivers
- Soldering iron, fine tip, temperature-controlled, set to 320°C
- Rosin-core solder, 0.5mm diameter
- Solder wick, fine gauge
- IPA (isopropyl alcohol 90% or better) and cotton swabs
- Multimeter (continuity and DC voltage)
- Fine-point tweezers
- Small flat-blade screwdriver for harness connectors

9.2 Pre-Installation Checks

1. Measure the output voltage of the power adapter at the barrel jack tip and ring before connecting to the machine. Confirm polarity. For Variant B, confirm the ideal diode bridge produces correct polarity output regardless of adapter polarity.
2. Test Q601 (2SB733) in diode mode: base-to-emitter and base-to-collector junctions should both read approximately 0.6V to 0.7V. A zero reading indicates the transistor has failed — replace with 2SA1015, BC327-40, or MMBT3906 before installing the module.
3. With the machine powered on batteries, measure the B+3 rail at the CP304 output. It should read approximately 10.8V. If zero or very low, CP304 has also failed — the module MT3608 boost stage will replace this function and no further action is required.
4. Measure the FG901 signal at IC601 pin 13 with a multimeter on AC voltage. A reading above zero confirms the FG circuit is operational. Note the approximate swing voltage for R3/R4 divider verification.

9.3 Installation Steps

5. Open the WM-D6C. Remove the four case screws (two each side). Separate upper and lower panels. Set the cassette mechanism aside.
6. Locate and unplug the CP304 harness connector from the main board. Remove the CP304 module from its cavity by releasing the retaining claw. Set aside.
7. Locate IC601 (CX20084) on the main board (position G-12). Using solder wick and iron, carefully remove IC601. Clean the pads with IPA. Identify pads 13 (FG signal), 14 (GND), 15 (motor drive output), and 16 (VDD) — these are the connection points for the module supplementary harness.
8. Locate X701 crystal and IC701 on the Auto-Off board. Desolder and remove both components. Clean pads with IPA.
9. For Variant A only: desolder CN301 from the lower panel. File or drill the panel opening to accept the USB-C panel connector. Fit the USB-C connector and secure with its retaining nut or adhesive as appropriate.

10. Connect the module's pre-assembled harness to J1 on the module board. Verify colour coding matches the harness diagram in the installation guide.
11. Route the supplementary harness wires (FG, motor drive, GND) to the IC601 pads on the main board. Solder each wire to its corresponding pad. Use minimum heat and minimum solder — these are fine pads on a 35-year-old board.
12. Plug the main harness connector into the original CP304 connector socket on the main board. The connector is keyed and will only insert one way.
13. For Variant B: connect the barrel jack wires from CN301 to the Variant B input terminals on the module.
14. Route the USB-C cable (Variant A) or verify the barrel jack is free to the panel (Variant B).
15. Reassemble the machine. Do not fully close the case — leave accessible for initial power-on test.
16. Connect power. Insert a test tape. Press play. Verify correct speed using the test tape tone and a frequency counter or spectrum analyser application on a phone.
17. Adjust RV601 (base speed trim) if necessary. RV602 (Speed Tune) should produce the expected speed shift across its travel. Verify Auto-Off function at end of tape.
18. Close the case. Installation complete.

10. Bill of Materials

All components are specified with LCSC part numbers for use with JLCPCB SMT assembly. Mouser and DigiKey equivalents are available for all parts. DNP (Do Not Populate) designators are noted for variant-specific components.

10.1 Shared Core Components

Ref Des	Part / Value	Package	Description
U1	STM32G0B1KBU6	UFQFPN32	Servo MCU — main processing element
U2	MT3608	SOT-23-6	6V → 10.8V boost converter for motor rail
U3	MCP1700T-3302E/TT	SOT-23-3	6V → 3.3V LDO for MCU supply
L1	4.7μH ±20% 1.5A	CDRH4D28-4R7NC or equiv	Boost inductor, max 3mm height
D_BOOST	SS14 1A 40V Schottky	SMA	Boost converter rectifier diode
D_FG1/2	BAT54 Schottky pair	SOD-123	FG signal clamp diodes × 2
C_VDD1-4	100nF 10V X5R	0402 × 4	STM32 VDD decoupling, one per pin
C_VDDA	100nF + 1μF 10V	0402 × 2	STM32 analog supply decoupling
C_VDDIO2	100nF + 4.7μF 10V	0402 × 2	STM32 VDDIO2 decoupling
C_LDO_IN	1μF + 10μF 10V	0402 + 0805	MCP1700 input decoupling
C_LDO_OUT	1μF + 10μF 10V	0402 + 0805	MCP1700 output decoupling
C_BOOST_OUT	22μF 16V	0805	Boost converter output capacitor
R3, R4	10kΩ, 22kΩ 1% 1/16W	0402 × 2	FG signal voltage divider
R5, R6	10kΩ, 100kΩ 5% 1/16W	0402 × 2	Option A motor drive (DNP if Option B)
R_ADC1-3	100Ω 5% 1/16W	0402 × 3	ADC input series protection
R_EN	10kΩ, 22kΩ 1% 1/16W	0402 × 2	Motor enable divider
J1	8-pin 1.25mm JST PH	Through-hole	WM-D6C machine interface connector
J_SWDD	4-pin 2.54mm header 1×4	Through-hole	SWD debug header
JP1	2-pin solder bridge	0402 pad	Option A/B mode select

10.2 Variant A Components — USB-C PD (DNP for Variant B)

Ref Des	Part / Value	Package	Description
U4	IP2721	SOT-23-6	USB PD sink trigger — 9V PDO request
U5	TPS62xxx (6V output)	SOT-23-5	9V → 6V synchronous buck regulator
U6	USBLC6-2SC6Y	SOT-23-6	USB D+/D− ESD protection array
J_USB	USB-C mid-mount receptacle	SMD	Power + data connector, panel mount variant
C_VBUS	4.7μF 16V X5R	0805	VBUS decoupling
C_BUCK_IN	10μF 16V X5R	0805	Buck converter input capacitor

C_BUCK_OUT	22μF 10V X5R	0805	Buck converter output capacitor
L_BUCK	4.7μH 1A	4040	Buck converter inductor
R_CC1/CC2	Determined by IP2721 CFG	0402 × 2	PD voltage selection resistors
R_VBUS_DIV	100kΩ + 47kΩ 1%	0402 × 2	VBUS voltage monitor divider to ADC

10.3 Variant B Components — Barrel Jack (DNP for Variant A)

Ref Des	Part / Value	Package	Description
U7	LTC4359 or MAX40200	SOT-23-5	Ideal diode bridge controller
Q_P1/2	Si2333DS P-ch MOSFET × 2	SOT-23	Ideal diode bridge FETs (positive path)
Q_P3/4	Si2333DS P-ch MOSFET × 2	SOT-23	Ideal diode bridge FETs (negative path)
F1	Polyfuse 500mA hold / 1.5A trip	0805	Resettable input overcurrent protection
D_TVS	SMBJ7.0A bidirectional TVS	SMA	Input overvoltage / transient clamp
C_BULK	220μF 10V electrolytic	5×5mm radial SMD	Input bulk storage capacitor
C_BYPASS	100nF 10V X5R	0402	High-frequency input bypass

10.4 Option B Motor Drive Components (DNP for Option A)

Ref Des	Part / Value	Package	Description
Q_LS	MMBT3904 NPN transistor	SOT-23	Level shift driver for Q601 base
R7	1kΩ 5% 1/16W	0402	Q_LS base series resistor
R8	10kΩ 5% 1/16W	0402	Q601 base drive resistor
R9	100kΩ 5% 1/16W	0402	Q601 base pullup to B+1 (motor off at boot)
C8	100nF 10V X5R	0402	PWM to analog RC filter capacitor

11. Repository Structure and Licensing

11.1 Repository Layout

```
wmd6c-servo/
├── hardware/
│   ├── kicad/           # KiCad 8+ schematic and PCB project
│   ├── gerbers/         # Production gerbers, versioned by release tag
│   ├── bom/             # Interactive HTML BOM, JLCPCB assembly CSV
│   └── datasheets/      # Referenced component datasheets
├── firmware/
│   ├── src/             # Bare-metal C source files
│   │   ├── main.c       # Startup, clock init, peripheral init, main loop
│   │   ├── servo.c      # PI control law, TIM2 ISR, DAC/PWM output
│   │   ├── adc.c        # ADC DMA scan for RV601/602/603
│   │   ├── usb_cdc.c    # USB CDC virtual COM port, command parser
│   │   ├── ucpd.c       # USB PD UCPD peripheral init and monitoring
│   │   └── flash.c      # Settings save/restore, last-sector wear levelling
│   ├── include/
│   │   └── config.h     # All tunable constants – edit here, rebuild
│   ├── build/
│   │   ├── Makefile     # arm-none-eabi-gcc, no IDE required
│   │   └── STM32G0B1KBUx_FLASH.ld # Linker script
│   └── releases/        # Pre-built .bin files for each tagged version
├── docs/
│   ├── installation/    # Step-by-step with photographs
│   ├── theory/          # Servo loop theory, signal chain analysis
│   ├── variants/        # Port notes for other compatible machines
│   └── troubleshooting/
├── CHANGELOG.md         # Human-readable change history per release
├── LICENSE_HW.txt       # CERN OHL-S v2
├── LICENSE_FW.txt       # MIT License
└── LICENSE_DOCS.txt     # CC BY-SA 4.0
```

11.2 Licensing

Hardware (KiCad files, gerbers, BOM)	CERN Open Hardware Licence Strong v2 (CERN OHL-S v2). Anyone distributing a modified version of the board must release their changes under the same licence. Commercial manufacture and sale of assembled boards is permitted.
Firmware (C source, Makefile, linker script)	MIT Licence. Permissive — anyone may use, modify, and distribute the firmware with or without the source, in commercial or non-commercial products, without releasing their modifications.
Documentation (this document and all docs/)	Creative Commons Attribution-ShareAlike 4.0 International (CC BY-SA 4.0). May be translated, adapted, and republished with attribution and under the same licence.

11.3 Contributing

Contributions of all kinds are welcomed and encouraged. The variant system is specifically designed to make machine-specific ports a bounded, well-defined task — a contributor who owns a WM-D3 or WM-D6 can add a variant directory under docs/variants/ with three artifacts: a schematic diff from the reference, a harness pinout diagram, and the calibrated FG target period for that machine's capstan speed. This constitutes a complete variant contribution without touching the core hardware or firmware.

Firmware contributions should maintain the zero-HAL, zero-RTOS architecture. Register-level C only. Any proposed abstraction that adds flash consumption or execution overhead requires explicit justification in the pull request.

TIP

The CHANGELOG.md uses plain English, not just git commit messages. Every resistor value change, every firmware constant update, and every layout revision is described in terms of what was observed on the bench and why the change was made. This discipline is non-optional — it is the institutional memory of the project.

12. Electrical Characteristics and Performance

12.1 Absolute Maximum Ratings

WARNING	Stresses above the values listed in this section may cause permanent damage to the module. These are stress ratings only and do not imply the module will function correctly at these limits.	
Power input voltage (Variant A, VBUS)	20V maximum	
Power input voltage (Variant B, barrel jack, either polarity)	+15V / -15V maximum	
B+1 output current	800mA maximum (derate above 600mA continuous)	
B+3 output current	700mA maximum peak	
Storage temperature	-40°C to +125°C	
Operating temperature	0°C to +60°C (module), -40°C to +85°C (STM32)	

12.2 Operating Characteristics

Parameter	Typical	Min	Max
B+1 output voltage	6.00V nominal	5.90V	6.10V
B+3 output voltage	10.8V nominal	10.5V	11.0V
VDD output voltage	3.30V nominal	3.27V	3.33V
MT3608 switching frequency	1.2 MHz	1.0 MHz	1.4 MHz
B+3 output ripple (playback)	<50mV p-p typical	—	80mV
Input voltage range (Variant A, VBUS)	9.0V PD nominal	8.5V	10.0V
Input voltage range (Variant B, post-bridge)	6.0V nominal	4.5V	9.0V
MCU supply current (64MHz, servo running)	~9mA typical	—	15mA
Total board quiescent current (no motor)	~50mA typical	—	80mA
Boot time to servo loop active	~2ms	—	5ms
FG input frequency range	500Hz to 10kHz	—	—
ADC resolution (pot reading)	12-bit (4096 steps)	—	—
DAC resolution (motor drive)	12-bit (4096 steps)	—	—
USB CDC baud rate (virtual)	Any (USB FS 12Mbit/s)	—	—

12.3 Expected Speed Performance

The module is designed to equal or exceed the original WM-D6C speed stability specification of 0.05% wow and flutter (WRMS). The digital PI servo loop offers several advantages over the original CX20084 analog implementation that contribute to this performance.

The target period is a fixed 32-bit integer stored in flash. It does not drift with temperature or supply voltage. The integral accumulator eliminates sustained steady-state speed error regardless of motor load variation or tape tension changes across the reel. The TIM2 input capture hardware timestamps FG edges with single-clock-cycle (15.6 nanosecond at 64 MHz) resolution, providing far finer period measurement than the original analog PLL could achieve. The servo loop executes on every FG edge — at a nominal FG rate of 2500 Hz, the control loop runs 2500 times per second, compared to the approximately 34 Hz effective update rate of the original crystal-divided reference.

Initial PI coefficient values ($K_p = 0.15$, $K_i = 0.008$) are starting points established by simulation. Final values must be tuned on the bench using the specific motor and mechanical assembly of each unit, using the test tape as a speed reference and the USB CDC telemetry output to observe the control loop behaviour. The optimised constants are then saved to flash and persist across power cycles.